THESIS METHODOLOGY | EVIDENCE REVIEW

Consolidated Research Review: Complaint Labels, Human Verification and Controlled Retraining

An evidence-based research brief for the proposed NLP - TF-IDF - SVM barangay complaint classification system

Research questions: How should Valid, Suspicious and Spam be labeled without confusing textual suitability with factual truth or administrative outcomes? What literature supports human review, annotation quality, synthetic-data limitations and controlled SVM retraining? What batch sizes should be investigated rather than assumed?

Prepared: 26 October 2026 | **Scope:** Three textual labels (Valid, Suspicious, Spam), administrator-reviewed feedback, synthetic training data and authentic complaint evaluation.

Bottom line. No reviewed paper establishes 25, 100 or 500 as a universally sufficient increment that guarantees improvement to an already-trained SVM. For this thesis, use cumulative research checkpoints of **+100, +250 and +500 verified examples** (and +1,000 only if feasible). Select an operational retraining trigger from the observed learning curve, class coverage, data quality, review cost and independent validation - not from an arbitrary number. [1-5,13]

How to use this report

Sections 1-5 retain the original batch-size, model-evaluation and retraining research. New Sections 6-8 incorporate the earlier classification-policy research, nine-case decision matrix, annotation study and record-design requirements. Section 9 consolidates the literature; Section 10 provides suggested manuscript language. Full clickable citations conclude the report.

Evidence labeling: "Published finding" means directly supported by the cited source. "Study design recommendation" is a proposal for this thesis that has not yet been validated on the user's complaint data. Example counts and performance scenarios are illustrative, not measured results.

1. Core definitions: what exactly is a batch?

Term	Meaning in this project
Labeling batch	Newly reviewed complaint narratives collected during a period or selected for human annotation.
Cumulative feedback count	Total **new eligible** examples added since the last reference training dataset was defined.
Retraining run	A deliberate fit of a candidate TF-IDF + SVM + probability-calibration pipeline using original training data plus the selected newly verified examples.
Model checkpoint	A comparison of a candidate model at a particular cumulative feedback count with the reference model, using unchanged validation criteria.
Deployment	Separately authorized replacement of the existing production pipeline after validation passes; it is not synonymous with retraining.

Do not confuse small annotation batches with small training sets

Let N0 represent the number of baseline training examples, and A represent the number of additional quality-checked labels. At a checkpoint, the candidate training size is **N0 + A**; the newly added complaints are not evaluated in isolation. A 500-example increment may be small relative to a very large baseline but disproportionately helpful if it covers missing real-world writing patterns. It may also contribute little if duplicates dominate. [1,3,4,13]

> **Important statistical distinction:** In Figueroa et al. [1], the reported 80-560 annotated samples are used in a study of learning-curve **prediction accuracy**; they are **not** evidence that retraining with 80-560 new complaints guarantees any particular increase in SVM F1.

What the evidence establishes

Published research	Relevant finding	What it does NOT establish
Figueroa et al. (2012) [1]	Learning curves can estimate needed labeled-data volume and uncertainty.	A universal retraining threshold of 100, 250 or 500.
Beatty et al. (2018) [2]	Small batches are often more label-efficient; larger batches can weaken stopping methods.	That a batch of 500 is always better than 100.
Ghose & Nguyen (2024) [3]	Active-learning benefits vary by dataset, representation, model, label budget and sampling strategy.	That uncertainty sampling always beats random selection.
Goudjil et al. (2018) [4]	A multi-class SVM can use informative human-labeled batches for text classification.	A barangay-specific sample-size rule.
Citovsky et al. (2021) [5]	Larger batches increase risk of redundant examples; diversity can help.	Direct validation of this thesis's SVM or complaint classes.

2. Recommended quantitative experiment

Use the following **cumulative checkpoints** as a preliminary study design. They represent increasing additions to the same baseline data, not three unrelated groups of complaints.

Model checkpoint	Total training examples	Interpretation / question
M0 - Baseline	N0	Existing fully fitted and calibrated NLP/TF-IDF/SVM pipeline.
M1 - +100	N0 + 100	Is there early evidence of a useful signal? Inspect errors; do not assume a deployable gain.
M2 - +250	N0 + 250	Does the additional verified feedback improve macro-F1, rare classes or false-Spam behavior?
M3 - +500	N0 + 500	Does the learning curve show a consistent gain sufficient to justify deployment?
Extension - +1,000	N0 + 1,000	Only if sufficient authentic quality-checked examples are available and earlier results indicate a benefit.

Example only: for a baseline of 10,000 training examples, the incremental additions +100, +250 and +500 are 1%, 2.5% and 5% of that baseline. The significance of these increases cannot be judged from percentage alone.

How many models are trained?

The core research comparison contains **one baseline plus three candidate checkpoints (four configurations)**. These configurations may each require multiple underlying fitting operations because cross-validation, calibration and tuning fit additional estimators. If you repeat experiments across seeds or compare random versus uncertainty/diversity sampling, the total computational fits increase; the number of deployable versions does not automatically increase.

How to sample the batches

Design requirement	Implementation
Nest the batches	Make the +100 examples a subset of +250 and +500, so the effect of accumulated feedback is interpretable.
Record provenance	Preserve source type, case ID, narrative version ID, reviewer ID and correction reason.
Limit duplicates	Review exact and near-duplicate complaints and template siblings; keep family groups in a single data partition.
Preserve classes	Report how many newly verified examples belong to Valid, Suspicious and Spam. Do not treat +500 as adequate if some classes have almost no new data.
Avoid selective feedback	Add an independently reviewed sample of predicted Valid cases to detect mistakes invisible to the normal Suspicious/Spam HITL queue. [8]

Study design recommendation: Treat +100, +250 and +500 as **research milestones**, not permanently hard-coded production thresholds. If no 500-example collection is feasible during the thesis timeframe, report the accessible batch sizes and associated uncertainty without inventing results.

3. Fair evaluation and acceptance criteria

Split once, then maintain isolation

Maintain clearly separated training and evaluation partitions; keep related complaint revisions and synthetic templates grouped. Fit the vocabulary, IDF weights, SVM and probability calibrator only on eligible training material, with disjoint calibration folds or data. Use a stable validation set for checkpoint selection and a separate untouched final test set for the final performance claim. [13-16]

The same validation set evaluates every checkpoint

Adding 100 examples does not mean evaluating on only 100. All candidates are assessed against the same independent validation examples with consistent labels and procedures. Where possible, compare performance on a second authentic-complaint-only slice so synthetic-text artifacts do not hide poor generalization. [6,13,14]

Primary / secondary metric	Reason it matters
Macro-F1 (primary)	Weights Valid, Suspicious and Spam equally, helping reveal a weak minority class.
Per-class precision, recall and F1	Shows which classes improve or deteriorate.
False-Spam-on-Valid count / rate	Safety-critical: understandable, actionable complaints should not be flagged as Spam.
Valid recall and confusion matrix	Detects misrouting of genuine clear narratives to HITL; catches class-specific regression.
Probability calibration and review-priority performance	Determines whether calibrated SVM scores are sufficiently reliable for confidence-guided review. [16]
Annotation and administrative workload	Quantifies reviewer burden per verified example and per improvement.

How much improvement is enough?

Specify acceptance criteria **before** looking at candidate scores. Recommended criteria: a meaningful improvement in the primary validation metric, no unacceptable deterioration in Valid recall or false-Spam performance, acceptable calibration, and no major regression on authentic-text evaluation. Use paired uncertainty estimates (e.g., bootstrap by independent case or family) to distinguish a small apparent improvement from sampling variation; these criteria must be operationalized using actual baseline errors.

> **No guaranteed gain:** It is entirely legitimate for the +500 candidate to perform worse than the baseline. In that situation, keep the existing model and document the failed candidate, its dataset changes and its evaluation. Do not repeatedly tune on the final test set. [3,14,17]

4. When to trigger retraining in the application

The application should count **eligible unique, quality-checked new text examples**, not clicks on Accept/Reject and not every repeated complaint. The administrator may review progress at a defined interval; a candidate fit is initiated only after an eligibility review and explicit authorization. The final number is selected after the thesis experiment, not assumed beforehand.

Condition	Recommended action
Insufficient verified feedback	Continue collecting; no model change.
Enough feedback but poor label diversity / excessive duplication	Improve labeling and selection; defer retraining.
Sufficient feedback, adequate class coverage and administrator authorization	Create a candidate training run.
Candidate fails predetermined validation criteria	Record results, retain existing deployed model, end this attempt.
Candidate passes criteria	Request independent deployment approval, version the entire pipeline and monitor rollback criteria.
Performance deteriorates before target count	Permit an exceptional evaluation or early authorized retraining review; do not automatically deploy.

Recommended system control specification

Display: total newly verified examples; eligible after quality checks; counts by verified class; corrected SVM predictions; percentage of near-duplicates; selected retraining threshold; last training run; deployed model version; candidate-validation summary. Provide distinct actions for **Review Data**, **Train Candidate** and **Approve Deployment**.

Compact workflow specification for the separate retraining diagram

Human-Reviewed Label Records Quality and Eligibility Checks Feedback Pool Retraining Justified and Approved?

No: Retain model; continue collecting.
Yes: Fit updated TF-IDF/SVM pipeline and recalibrate Compare against deployment criteria **Pass:** approve/version/deploy; **Fail:** retain deployed model and archive candidate. [14,16,17]

Do not call every batch update "active learning"

A pool of labeled HITL reviews followed by retraining is accurately called **human-feedback-driven batch retraining**. The stronger term **active learning** is justified if the system actually uses a defined sample-selection or prioritization strategy to request informative annotations and evaluates it (for example, uncertainty plus diversity versus a random-audit baseline). [3-5,12]

5. Addressing synthetic data and label ambiguity

Why authentic verified narratives are unusually valuable

Li et al. [6] found that the effectiveness of LLM-generated synthetic text is inconsistent across tasks and declines as classification subjectivity increases. The implications for a barangay dataset are serious: polished synthetic Valid narratives and templated Spam may allow the SVM to learn generation patterns rather than how actual residents write. New authentic, quality-reviewed examples are valuable mainly because they test and enrich the model's coverage - not merely because they increase N.

Keep textual labels separate from case outcomes

Model / administrative field	Recommended interpretation
Predicted class: Valid / Suspicious / Spam	SVM estimate from the complaint narrative.
Verified text label	Human assessment of the original or revised narrative under the written annotation policy. [7]
Accepted / Pending Clarification / Rejected	Administrative disposition; **not** an automatic ML ground-truth label.
Unresolved / Ongoing / Resolved	Statuses within separate Case Management for accepted complaints.

Illustrative edge case: an original "May problema dito, pakitingnan" may remain Suspicious because it lacks an identifiable issue. A later clarified complaint can independently be Valid, and the case can be Accepted. Training on the original sentence as Valid solely because the reported incident was eventually confirmed would contradict the labeling rule and introduce information unavailable to the text model at submission time. [7,8,14]

Recommended annotation and selection safeguards

• **Explicit definitions:** Valid = understandable and actionable text, not established factual truth; Suspicious = materially incomplete, ambiguous or inconsistent complaint-related text; Spam = clearly non-complaint, promotional or irrelevant submissions. [7]

• **Quality review:** Sample cases from every class; adjudicate disagreement and preserve label-correction rationale. [7]

• **Sample selection:** Balance uncertain cases with diversity controls and random audits of predicted Valid examples, to mitigate selective labels and redundancy. [3,5,8]

• **Privacy:** Store sensitive originals securely; create appropriately minimized research copies and follow transparency, legitimate-purpose and proportionality principles. [18]

• **Human decision:** Suspected Spam is not proof of Spam. Avoid automatic rejection in the initial system because the cost of false positives can be high. [9-11,19]

6. Operational labels: a defensible three-class ontology

The earlier research discussion proposed a **text-suitability taxonomy** rather than a truth-detection taxonomy. Röttger et al. [7] motivate choosing explicit prescriptive annotation rules; Fleisig et al. [20] highlight the value of identifying genuine annotator disagreement rather than hiding it. **No cited paper validates this precise three-class taxonomy for Philippine barangay complaints.** These operational definitions are proposed and must be evaluated.

Class	Proposed operational definition	Exclusion / safeguard
Valid	A coherent, complaint-related narrative with enough understandable information to identify and process the issue.	Does **not** mean the reported incident has been factually proven.
Suspicious	Complaint-related text materially incomplete, ambiguous or internally inconsistent, preventing reliable interpretation or processing without clarification.	Does **not** mean dishonesty, fabrication or malicious intent.
Spam	Clearly non-complaint or irrelevant submissions, such as unsolicited advertising, promotional material or similarly identifiable spam content.	Does **not** include genuine reports merely because they are repeated, emotional, short, misspelled or unverified.

Three dimensions are distinct

Dimension	Possible values	What it records
Text prediction	Valid / Suspicious / Spam	What the currently deployed classifier inferred from the available narrative.
Administrator-verified text label	Valid / Suspicious / Spam	Reviewer-confirmed classification of that **particular text version** using the annotation manual.
Administrative disposition	Accepted / Pending Clarification / Rejected	Whether the **submission** proceeds; not a training label by itself.
Case-management status	Unresolved / Ongoing / Resolved	Progress of an accepted case in its separate workflow.
Incident verification	Confirmed / Unverified / Disputed / Other documented outcome	Evidence-based administrative finding about the reported event, **not** inferred from SVM text features.

Central rule: A complaint may be **Suspicious as original text and Accepted administratively**. Conversely, a clear Valid narrative may later be administratively rejected for a documented reason. Neither outcome automatically changes the original training label. [7,20,21]

Why truth must not be conflated with narrative quality

Irnawan et al. [21] study claim-veracity detection using evidentiary support in a **different domain** (fake-news datasets). The relevant transferable idea is that assessing whether a claim is true requires evidence beyond its wording. This paper does **not** establish the factual accuracy of local complaint narratives or validate the proposed class taxonomy.

7. Complete nine-scenario decision matrix

The following matrix retains all nine difficult cases from the earlier discussion. The label refers to the **version of text under assessment**; disposition is an independent administrative decision. These are **proposed** application and annotation policies, not findings measured in the thesis dataset. [7,20,21]

Case requiring careful handling	Verified text label	Recommended administrative handling
1. SVM predicts Suspicious, yet the **original** narrative is clear and actionable.	**Valid** - correct SVM error.	Accept and route to separate Case Management.
2. Original is incomplete; resident supplies a sufficiently clear revised narrative.	Original **Suspicious**; revised **Valid**.	Accept the clarified complaint; preserve both versions.
3. Original is incomplete; external evidence supports accepting without rewriting it.	Original remains **Suspicious**.	May Accept if the approved administrative policy allows it.
4. SVM predicts Spam, but original text is a genuine, coherent complaint narrative.	**Valid** - correct false Spam prediction.	Accept and route for case processing.
5. SVM predicts Spam, but text is complaint-related and lacks essential details.	**Suspicious** - correct false Spam prediction.	Request clarification; may Accept when sufficient external information exists.
6. SVM predicts Suspicious, but the text is clearly promotional or irrelevant.	**Spam** - correct false Suspicious prediction.	Reject after human confirmation; record reason.
7. Text is clear, but the factual allegation cannot be substantiated.	**Valid** as text.	Investigate appropriately; no automatic Spam relabeling.
8. Text is clear, but later administrative evidence establishes grounds for rejection.	**Valid** as text.	Reject or refer according to documented grounds, not because the text became Spam.
9. Several residents report the same problem, or versions are repeated.	Independently assess each narrative version.	Corroborate or link cases; check duplicates. Repetition alone is not Spam.

Reject option in ML is not administrative rejection

Franc, Prusa and Voracek [22] develop classifiers that can **abstain** from prediction, sometimes called a machine-learning reject option. Here, an administrative **Rejected** submission means a human decision about case eligibility. The two concepts should be distinguished throughout the manuscript.

Risk of incorrect label propagation: Do not convert all Accepted submissions into Valid training examples or all administratively Rejected submissions into Spam examples. Those shortcuts use outcomes that the text-only classifier could not observe at submission time. [7,8,20]

8. Annotation study, data records and HITL handoff

Independent annotation pilot - study design recommendation

Before generating or labeling the full training dataset, draft an annotation manual with definitions, positive/negative examples and exceptions. Use **at least two trained reviewers** for a representative pilot sample; this is a **recommended study protocol**, not a minimum mandated by Röttger or Fleisig. Compare their labels, report agreement/disagreement, and adjudicate contested examples with documented reasons. Reviewers should label the presented narrative, not guess unseen facts about whether an incident occurred. [7,20]

Pilot strata should deliberately include missing locations, incomplete issue descriptions, conflicting dates, code-switched Filipino/English narratives, dialect variation, misspellings, emotional or repeated complaints, advertisements, near-duplicate submissions, false allegations and unclear spam-like phrasing. Inspect which ambiguities are intrinsic to the text versus caused by an unclear written rule. [7,20]

Example of version-preserving annotation

Research / system field	Illustrative entry
Case identifier / split group	Case 021 / group-021 (all versions held in same data split)
Original narrative	"May problema dito, pakitingnan."
Initial SVM prediction	Suspicious
Example predicted confidence	0.71 (hypothetical)
Verified original-text label	Suspicious - insufficient issue description
Clarified narrative	"Sirang streetlight sa Purok 3."
Verified revised-text label	Valid - now actionable
Administrative disposition	Accepted
Separate case status	Unresolved (then Ongoing / Resolved if processed)
Training eligibility	Original and revised versions separately quality-checked; no leakage between dataset partitions

HITL referral rule and the two outputs

Under the agreed **current project policy**, predictions of **Valid** go directly to Case Management, while predictions of **Suspicious or Spam** enter the normal administrator HITL queue. Calibrated confidence can prioritize reviews **inside that queue**. This is a class-based routing rule, **not** a claim that all low-confidence Valid predictions are reviewed. A separate representative audit of predicted Valid cases is important for detecting unobserved errors. [8,11,16,22]

End of HITL	Destination / meaning
Accepted submission	To separate Complaint Case Management; original text label remains unchanged unless explicitly corrected.
Pending Clarification	Await additional information; original and revised narratives retain distinct versioned labels.
Rejected / administratively closed	Record status and reason; do not automatically learn Spam from the disposition.

End of HITL	Destination / meaning
Reviewed label record	To label-quality control; only eligible, nonduplicate, privacy-conscious verified text examples enter feedback training pool.

Minimum traceability and privacy metadata

Retain complaint and narrative-version IDs, original SVM class and calibrated probabilities, reviewer identifiers, verified text labels, correction reasons, administrative dispositions, supporting evidence references, review timestamps, source type (synthetic/authentic), split-family IDs and training eligibility. Protect personal information and use a minimized training representation. [7,8,18,20]

9. Evidence synthesis and explicit research limitations

Research paper	Defensible insight	What it does not establish
Röttger et al. (2022) [7]	Choose explicit annotation paradigms and consistently applied labels.	These exact barangay Valid/Suspicious/Spam definitions.
Fleisig et al. (2024) [20]	Annotator disagreement can contain useful information.	That every contested label should be settled by a particular voting rule.
Li et al. (2023) [6]	Synthetic text performance varies and can suffer with subjective categories.	That an LLM-generated barangay corpus transfers perfectly to actual complaints.
Irnawan et al. (2025) [21]	Factual-veracity claims benefit from evidence beyond text.	That text-only SVM can establish a barangay incident as true or false.
Franc et al. (2023) [22]	A classifier can abstain when uncertain.	That ML abstention legally equals rejection of a public complaint.
Lakkaraju et al. (2017) [8]	Selective observed outcomes can bias model evaluation.	That reviewing only Suspicious/Spam is representative of the full complaint stream.
Figueroa et al. (2012) [1]	Learning curves inform sample-size planning.	That adding 100 or 500 complaints guarantees model gains.

Explicit findings versus thesis hypotheses

• **Published:** Under some tasks, synthetic-data performance is inconsistent [6]; subjective annotation requires explicit policy [7,20]; abstention and human deferral are useful design families [10,11,22].

• **Proposed:** The three textual classes, class-based HITL routing, manual disposition actions, +100/+250/+500 checkpoints and controlled deployment are methodological choices for this thesis.

• **To test:** How often predicted Spam is confirmed, how often Suspicious is corrected to Valid, reviewer agreement, and whether authentic feedback improves macro-F1 and reduces false-Spam errors.

• **Limitations:** Without authentic reviewed evaluation data, performance conclusions are limited to available test distributions. Repeated reports are not proof of truth; limited narrative text cannot establish incident veracity. [6,21]

Consolidated research position for manuscript discussion

> The proposed SVM classifier identifies the textual suitability of barangay complaint submissions using three operational categories: **Valid, Suspicious and Spam**. These categories do not independently establish the factual truth of reported incidents. Predictions requiring verification are reviewed by authorized administrators, who assess the original narrative, supporting information and applicable complaint-handling criteria. Human reviewers may confirm or correct the model text label while recording administrative acceptance or rejection separately. **Quality-checked text labels, rather than case outcomes alone**, support subsequent feedback-driven model retraining. [7,8,20-22]

Misinterpretations to avoid: No paper establishes a universal 500-case retraining minimum [1-5]; more synthetic or repeated complaints do not necessarily increase F1 [3,5,6]; Accepted is not automatically Valid [7,20]; an SVM decision score is not a calibrated probability [16]; and the ML reject option is not rejection of the citizen submission [22].

10. Manuscript-ready suggested Methodology wording

Operational class definitions and human verification. The proposed classifier assigns text-based labels of Valid, Suspicious or Spam under an explicit annotation manual. These labels describe narrative clarity and relevance rather than the factual accuracy of the allegation. Predicted Suspicious and Spam complaints enter administrator verification; predicted Valid submissions follow the normal case-management route under the selected project policy. Human reviewers confirm or correct labels for the specific narrative version being reviewed, whereas administrative dispositions of Accepted, Pending Clarification or Rejected are recorded independently. Original incomplete text can retain Suspicious while a clarified revision becomes Valid, and repeated acceptance does not itself alter a training label. This labeling design is informed by research on prescriptive annotation, disagreements, veracity and selective labels. [7,8,20,21]

Annotation pilot and data quality. A representative sample of synthetic and available authentic complaint narratives will be independently annotated by trained reviewers using the project annotation manual. Reviewer disagreements will be recorded and adjudicated; outcomes will be used to refine class boundaries before final dataset generation and evaluation. Human-reviewed records will be versioned and screened for privacy, duplicates, label quality and train-test leakage before entering any retraining pool. This describes the **proposed** protocol and should be stated as implemented only when completed. [6-8,18,20]

Retraining data accumulation and candidate selection. The proposed system adopts a controlled, human-feedback-driven model improvement procedure rather than retraining after each reviewed complaint. Administrator-reviewed narratives are retained with their verified textual labels, source-version identifiers and audit records. Before a feedback example is added to the eligible training pool, it is screened for label consistency, duplication, relevance and privacy-related risks. This design responds to documented concerns regarding subjective annotation, selective-label bias and the variable utility of synthetic text for classification. [6-8,18]

Batch-size investigation. The study evaluates cumulative increments of 100, 250 and 500 newly verified narratives as provisional learning-curve checkpoints, with an extension to 1,000 if the quantity and quality of available data permit. At each checkpoint, the candidate TF-IDF-SVM model is refitted using the baseline training data plus the eligible new examples; probability calibration is conducted without evaluation leakage. These counts are experimental research choices, not assumed minimum requirements. Published evidence indicates that sample-size requirements and active-learning efficiency vary across datasets and sampling conditions. [1-5,13-16]

Model comparison and release decision. Candidates are compared using unchanged validation protocols, emphasizing macro-F1, per-class performance, erroneous Spam predictions of Valid narratives, calibration and reviewer workload. Versions of the same complaint and related synthetic templates are kept within the same split. A candidate is deployable only after meeting predefined acceptance criteria and receiving authorization. Otherwise, the currently deployed pipeline remains operational and the failed candidate is documented for analysis. An independent final test set is reserved for the selected model's final evaluation. [13,14,16,17]

Do not present the above as completed results. Replace "proposed" with "implemented" only if the corresponding software, feedback collection, batch-size experiments and validation outputs actually exist. Replace the provisional 100/250/500 checkpoints with empirically selected values if later research findings warrant a change.

Recommended implementation decision

Thesis experiment: baseline plus +100 / +250 / +500 cumulative candidate checkpoints. **Operational application:** show count and class composition of eligible verified examples, let authorized personnel request candidate retraining when criteria are satisfied, and deploy only after passing evaluation. **Do not hard-code 500 as a guarantee of improvement.**

References (primary publications and official sources)

Numbered citations in the report refer to the sources below. Links are clickable. Peer-reviewed papers and official technical/regulatory material are distinguished by their bibliographic descriptions; research claims have not been extrapolated as universal barangay-specific findings.

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